

Gradient-based dimension reduction

Part 1: Curse of dimension

Olivier Zahm

`olivier.zahm@inria.fr`

INRIA – Université Grenoble-Alpes
<https://team.inria.fr/airsea/en/>

PILearnWater

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Overview

- ▶ **Part 0:** Curse of dimension
 - **Sensitivity Analysis** for newbies
 - **Curse of dimensionality** in function approximation
 - Illustration on **Kernel Ridge Regression**
 - Why gradients?
- ▶ **Part 1:** Linear dimension reduction for regression problems
 - Active Subspace
 - Poincaré inequalities
 - Towards non-linear dimension reduction
- ▶ **Part 2:** Linear dimension reduction for inverse problems
 - Likelihood-informed subspace
 - Logarithmic Sobolev inequalities and co
 - Towards Optimal Experimental Design

Context

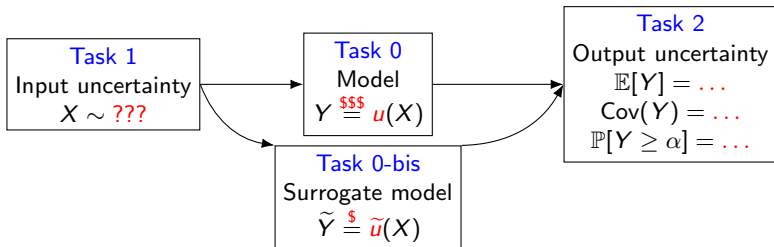
$$Y = u(X_1, \dots, X_d)$$

- ▶ $X \sim \pi_X$: input parameter, typically with **product** density $\pi_X(x)$
- ▶ Y : output of interest, generally **scalar** $Y \in \mathbb{R}$
- ▶ u : **expensive** numerical model

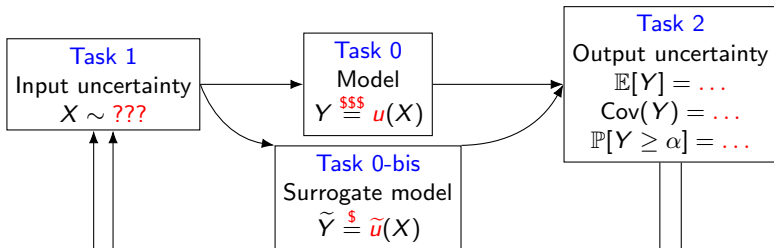
Example: $u(X) = \int v d\Omega$, where v solved the **parametrized PDE**

$$-\operatorname{div}(\kappa \nabla v) = f \quad \text{on } \Omega, \quad \text{where} \quad \kappa = \kappa(X),$$

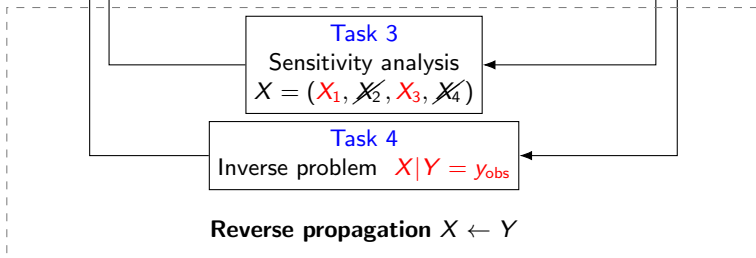
Uncertainty propagation $X \rightarrow Y$

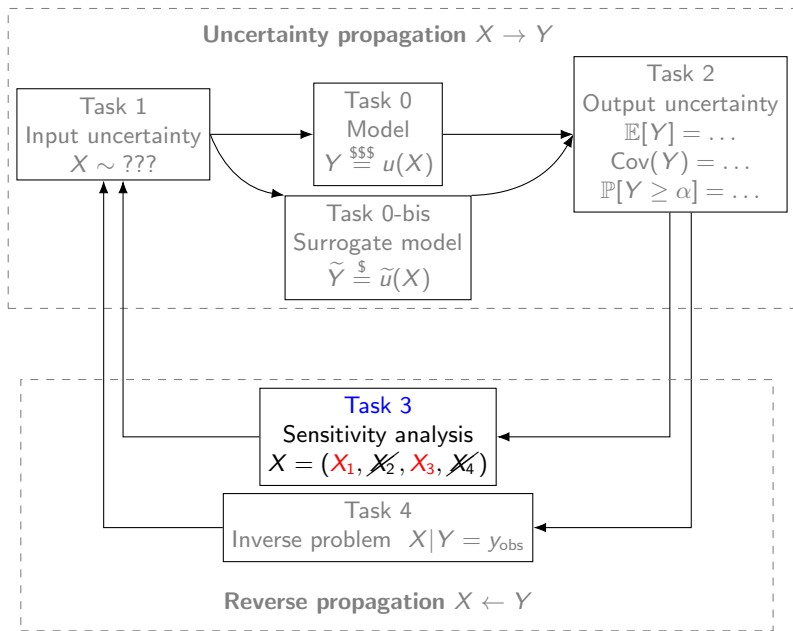


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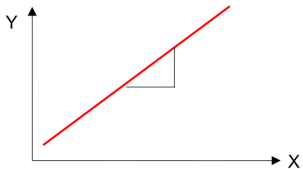


Reverse propagation $X \leftarrow Y$

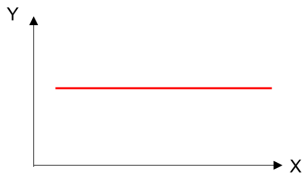




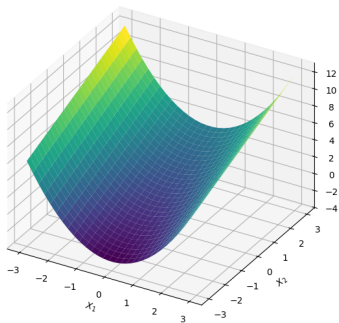
Part 0: Variance-based **sensitivity analysis**
for newbies (10 min max)



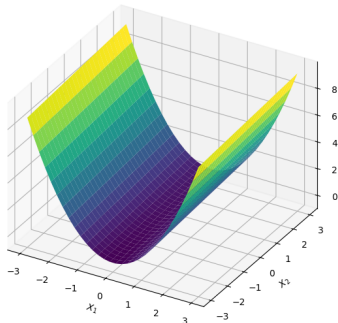
Y is **sensitive** to X



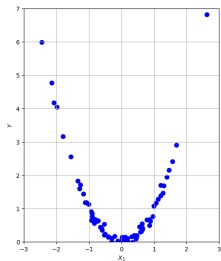
Y is **not sensitive** to X



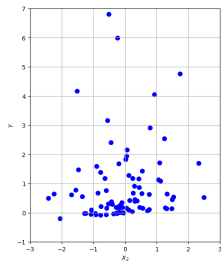
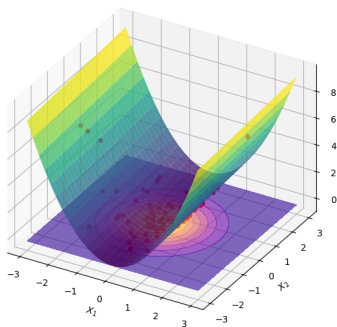
Y is **sensitive** to both X_1 and X_2



Y is **not sensitive** to X_2

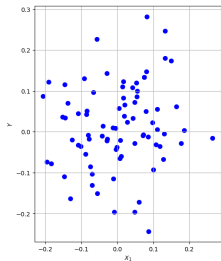


Y V.S. X_1

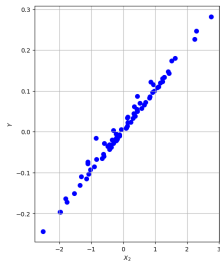
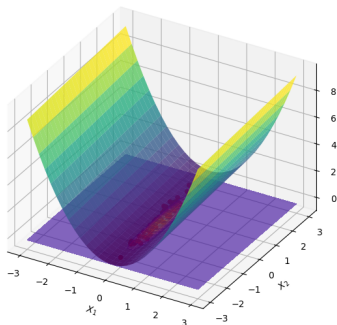


Y V.S. X_2

Y can be explained by X_1 , whereas Y seems independent on X_2



Y V.S. X_1



Y V.S. X_2

~~Y can be explained by X_1 , whereas Y seems independent on X_2~~

If we change the law π_X of X , the conclusion can be reversed.

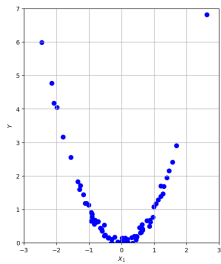
Today, we focus on **Sobol' indices**



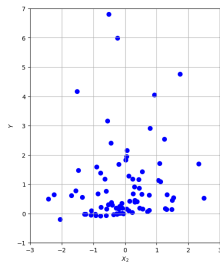
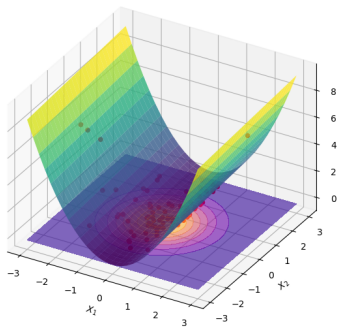
Ilya Sobol



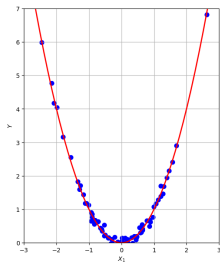
[Iooss, Da Veiga, Prieur, Gamboa 2021: Basics and trends in SA: theory and practice in R]



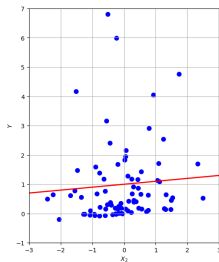
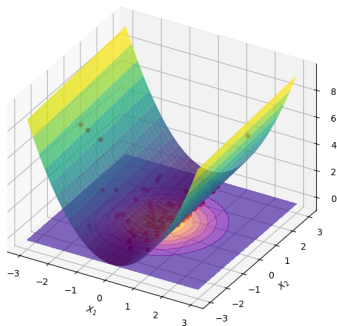
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Y V.S. X_2



Y V.S. X_1

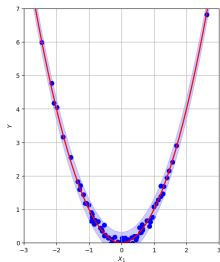


Y V.S. X_2

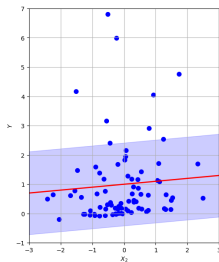
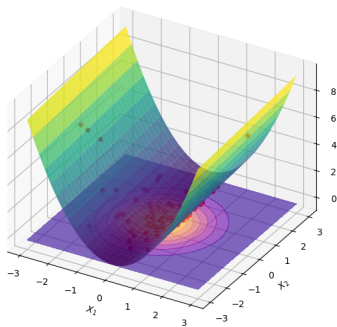
For $\tau \subset \{1, \dots, d\}$, let $X_\tau = (X_i)_{i \in \tau}$ and $X_{-\tau} = (X_i)_{i \notin \tau}$

- **Conditional expectation:** if the variance of $\mathbb{E}[u(X)|X_\tau]$ is close to the variance of Y , then X_τ is an explanatory factor. **Closed Sobol' index:**

$$S_\tau = \frac{\text{Var}(\mathbb{E}[u(X)|X_\tau])}{\text{Var}(u(X))} \approx 1 \quad \Rightarrow \quad X_\tau = \text{explanatory factor}$$



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Y V.S. X_2

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- **Conditional variance:** if $\text{Var}(u(X)|X_1) \ll \text{Var}(Y)$, then $u(X_1^{\text{freeze}}, X_2)$ doesn't recover the variance of $Y = u(X)$. **Total Sobol' index:**

$$T_\tau = \frac{\mathbb{E}[\text{Var}(u(X)|X_{-\tau})]}{\text{Var}(u(X))} \approx 0 \quad \Rightarrow \quad X_\tau = \text{irrelevant factor}$$

- ▶ There exists **many statistical estimators** for S_τ and T_τ !
- ▶ There exists **many other variance-based** indices!
- ▶ There exists **many other non-variance-based** indices!

My (favorite) perspective is the function approximation in $L^2_{\pi_X}$:

$$\min_{f: X \mapsto \mathbb{R}} \|u - f\|^2 =: \mathbb{E}[\text{Var}(u(X)|X_\tau)]$$

Sobol' indices: the **function approximation** perspective

Using the **total variance formula (*)**, the **closed Sobol' indices** writes

$$S_{\tau}(u) := \frac{\text{Var}(\mathbb{E}[u(X)|X_{\tau}])}{\text{Var}(u(X))} \stackrel{(*)}{=} 1 - \frac{\min_{f: X \mapsto f(X_{\tau})} \|u - f\|^2}{\text{Var}(u(X))}$$

$\begin{aligned} S_{\tau}(u) \approx 1 &\Leftrightarrow u(X) \approx f(X_{\tau}) \\ &\Leftrightarrow X_{\tau} \text{ "explains" well } Y = u(X) \\ &\text{usefull for ranking the } X_{\tau} \end{aligned}$

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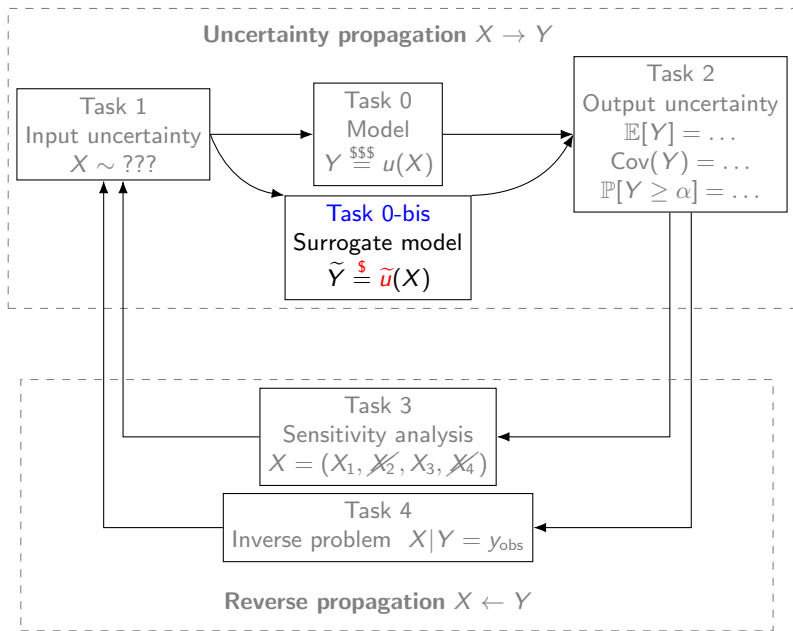
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Similarly, the **total Sobol' indices** writes

$$T_\tau(u) := 1 - \frac{\text{Var}(\mathbb{E}[u(X)|X_{-\tau}])}{\text{Var}(u(X))} \stackrel{(*)}{=} \frac{\min_{f: X \mapsto f(X_{-\tau})} \|u - f\|^2}{\text{Var}(u(X))}$$

$$\begin{aligned} T_\tau(u) \approx 0 &\Leftrightarrow u(X) \approx f(X_{-\tau}) \\ &\Leftrightarrow X_\tau \text{ is useless to "explain" } Y = u(X) \\ &\text{usefull for screening the } X_\tau \end{aligned}$$



Part I: The curse of dimensionality in function approximation

Curse of dimensionality [Novak and Woźniakowski 2009]

Theorem (informal): no algorithm can approximate any smooth function
 $u : [0, 1]^d \rightarrow \mathbb{R}$ with less than $2^{d/2}$ point evaluations of u .

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► **Sparse structure**

$$u(x) \approx \sum_{\alpha \in \Lambda_N} u_{\alpha} \Phi_{\alpha}(x), \quad \begin{cases} \Phi_{\alpha}(x) : \text{well-chosen basis} \\ \#\Lambda_N = N \end{cases}$$

► **Low-rank structure**

$$u(x) \approx \sum_{i=1}^r u_1^i(x_1) \dots u_d^i(x_d)$$

► **Compositional structure**

$$u(x) \approx f \circ g(x), \quad \begin{cases} g : \mathbb{R}^d \rightarrow \mathbb{R}^m \\ f : \mathbb{R}^m \rightarrow \mathbb{R} \\ m : \text{latent dimension} \end{cases}$$

► ...

A prototypical example: parametrized elliptic PDE

Find $u(x) \in H^1(\Omega)$ solution to

$$-\operatorname{div}(a(x) \nabla u(x)) = f \quad \text{in } \Omega \subset \mathbb{R}^2 \text{ or } \mathbb{R}^3$$

where

$$a(x, s) = a_0(s) + \sum_{i=1}^{d=\infty} x_i a_i(s), \quad \begin{cases} x_1, x_2, \dots \in [-1, 1] \\ s \in \Omega \end{cases}$$

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Assume

$$(\|a_i\|_\infty)_{i \geq 1} \in \ell^p \quad \text{for some } p < 1$$

then there exists

$$\tilde{u}(x, s) = \sum_{i=1}^n \varphi_i(x) v_i(s), \quad \begin{cases} v_i \in H^1(\Omega) \\ \varphi_i : \text{polynomials} \end{cases}$$

such that

$$\sup_{x \in [-1, 1]^N} \|u(x) - \tilde{u}(x)\|_{H^1(\Omega)} \leq C n^{-s}, \quad \text{where } s = p^{-1} - 1 > 0$$

for some constant C **independent on the dimension** (no curse of dimensionality!)



[Cohen DeVore 2015: Approximation of high-dimensional parametric PDEs, Acta Numerica]

A prototypical example: parametrized elliptic PDE

Near-optimal approximations can be obtained using

- **sparse polynomial** expansions:

$$\tilde{u}(x, s) = \sum_{\substack{\alpha \in \Lambda_n \\ \#\Lambda_n = n}} \varphi_\alpha(x) v_\alpha(s) \quad \left\{ \begin{array}{l} \varphi_\alpha(x) : \text{given multivariate polynomials} \\ \Lambda_n : \text{Greedy algorithm } \Lambda_{n+1} = \Lambda_n \cup \{\alpha_{n+1}^*\} \\ v_\alpha(s) : \text{least-squares, interpolation, ...} \end{array} \right.$$

- the **Reduced Basis** method:

$$\tilde{u}(x, s) = \sum_{i=1}^n \varphi_i(x) u(x_i, s) \quad \left\{ \begin{array}{l} x_1, \dots, x_n : \text{Greedy algorithm } n \leftarrow n + 1 \\ \varphi_i(x) : \text{Galerkin projection of } u(x) \text{ on} \\ \quad \text{span}(u(x_1), \dots, u(x_n)) \end{array} \right.$$

-  [Rozza, Huynh and Patera 2008] *Reduced basis approximation and a posteriori error estimation for affinely parametrized elliptic coercive PDE* Arch. Comput. Methods Eng.
-  [Blatman and Sudret 2011] *Adaptive sparse polynomial chaos expansion based on least angle regression*
-  [Chkifa, Cohen and Schwab 2015] *Breaking the curse of dimensionality in sparse polynomial approximation of parametric PDEs* Journal de Mathématiques Pures et Appliquées
-  [.....]

Illustration of the curse of dimension with the **Kernel Ridge Regression** method

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Let us make the (strong!) assumption that u belongs to a **Reproducing Kernel Hilbert Space (RKHS)**

$$u \in \mathcal{H} = \left\{ f : \mathbb{R}^d \rightarrow \mathbb{R}, \|f\|_{\mathcal{H}}^2 = \langle f, f \rangle_{\mathcal{H}} < \infty \right\}$$

s.t. $\mathcal{H} \ni f \mapsto f(x) \in \mathbb{R}$ **continuous** $\forall x$,

Notably, the Sobolev space $H^1([0, 1])$ is a RKHS, but $H^1([0, 1]^2)$ is not...

RKHS are characterized by their **Kernel** $\kappa = \kappa(\mathcal{H})$

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Riesz representation theorem: there exists a **feature map** $\varphi : x \mapsto \varphi(x) \in \mathcal{H}$ such that

$$f(x) = \langle f, \varphi(x) \rangle_{\mathcal{H}}$$

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Define the **kernel** $\kappa : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ by

$$\kappa(x, x') := \langle \varphi(x), \varphi(x') \rangle_{\mathcal{H}}$$

which is a **symmetric positive definite (SPD) function**, i.e. $\forall x_1, x_2, \dots \subset \mathbb{R}^d :$

$$[\kappa(x_i, x_j)]_{ij} \succeq 0$$

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Since $\kappa(x, \cdot) = \langle \varphi(x), \varphi(\cdot) \rangle_{\mathcal{H}} = \varphi(x)(\cdot)$, we deduce the **reproducing property**

$$f(x) = \langle f, \kappa(x, \cdot) \rangle_{\mathcal{H}}$$

so that

$$\mathcal{H} = \left\{ f(x) = \sum_{i=1}^{\infty} \alpha_i \kappa(x, x_i) \left| \begin{array}{l} x_i \in \mathbb{R}^d \text{ and } \alpha_1, \alpha_2, \dots \in \mathbb{R} \text{ s.t.} \\ \|f\|_{\mathcal{H}}^2 := \sum_{i,j=1}^{\infty} \alpha_i \alpha_j \kappa(x_i, x_j) < \infty \end{array} \right. \right\}$$

Observe that κ entirely characterizes $\mathcal{H} = \mathcal{H}(\kappa)$.

Choosing the kernel

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$$\kappa(x, x') = \exp\left(-\frac{1}{2\ell^2} \|x - x'\|^2\right)$$

for some bandwidth $\ell > 0$, classically $\ell = \text{med}(\|X^{(i)} - X^{(j)}\|)$.

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- ▶ Matérn kernels $\kappa_\nu(x, x')$ with regularity parameter $\nu > 0$

(non-differentiable) $\kappa_{1/2}(x, x') = \exp\left(-\frac{\|x - x'\|}{\ell}\right)$

(once differentiable) $\kappa_{3/2}(x, x') = \left(1 + \frac{\sqrt{3}\|x - x'\|}{\ell}\right) \exp\left(-\frac{\sqrt{3}\|x - x'\|}{\ell}\right)$

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- ▶ Take your favorite (finite) basis function $\Phi(x) = (\Phi_1(x), \dots, \Phi_K(x))$, e.g. polynomial, wavelet, Fourier, spline,..., and define

$$\kappa(x, x') = \Phi(x)^\top \Phi(x')$$

In this case, $\dim(\mathcal{H}) \leq K$

Kernel ridge regression: $\min \mathbb{E}[(Y - f(X))^2]$, $f \in \mathcal{H} \subset L^2_{\pi_X}$

Given i.i.d. samples $\{(X^{(i)}, Y^{(i)})\}_{i=1}^N$ with $Y^{(i)} = u(X^{(i)})$, consider the regularized¹ problem

$$\min_{f \in \mathcal{H}} \sum_{i=1}^d \left(Y^{(i)} - f(X^{(i)}) \right)^2 + \lambda \|f\|_{\mathcal{H}}^2$$

¹“Ridge” or “Tikhonov” regularization: λ controls the complexity of the solution

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By the [representer theorem](#), the solution necessarily writes

$$f^N(x) = \sum_{i=1}^N \alpha_i \kappa(x, X^{(i)}), \quad \alpha \in \mathbb{R}^N$$

and the coefficients $\alpha = (\alpha_1, \dots, \alpha_N)$ are obtained via [least-square regression](#)

$$\alpha = (K + \lambda I)^{-1} Y_N \quad \text{where} \quad \begin{cases} K = [\kappa(X^{(i)}, X^{(j)})]_{i,j} \\ Y_N = (Y^{(1)}, \dots, Y^{(N)}) \end{cases}$$

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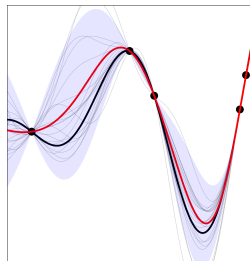
Link with Gaussian Process (*i.e.* Kriging)

KRR	GP
$u \in \mathcal{H}$	u is a realization of $Z \sim \mathcal{N}(0, \kappa)$
$\kappa = \text{kernel}$	$\kappa(x, x') = \text{Cov}(Z(x), Z(x'))$
$f^N = \text{LS problem}$	$f^N(x) = \mathbb{E}[Z(X) Z(X^{(i)}) = Y^{(i)}]$
$\lambda = \text{regularization}$	$\lambda = \text{nugget}$

The **little extra** with GP is the notion of *posterior law*

$$Z(x) | Z(X^{(i)}) = Y^{(i)}$$

posterior covariance, posterior samples...



See how vanilla KDR **fails** in high dimensions:



`gkdr_demo.ipynb`

Accounting for **anisotropy** in the kernel

Consider the modified Gaussian RBF

$$\kappa_B(x, x') = \exp \left(-\frac{1}{2\ell^2} \|B^\top (x - x')\|^2 \right)$$

for some matrix $B \in \mathbb{R}^{d \times m}$ to determine...

$f \in \mathcal{H}_{\kappa_B} \quad \Rightarrow \quad f(x) = f(B^\top x)$

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►  [Fukumizu 2004]: optimize over $B \in \mathbb{R}^{d \times m}$

$$\text{“} \min_B \left\{ \min_{f \in \mathcal{H}_{\kappa_B}} \sum_{i=1}^d (Y^{(i)} - f(X^{(i)}))^2 + \lambda \|f\|_{\mathcal{H}_B}^2 \right\} \text{”}$$

Hard to solve numerically... (many local minima, expensive gradients)

Accounting for **anisotropy** in the kernel

Consider the modified Gaussian RBF

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$$H = \mathbb{E}[\nabla u(X) \nabla u(X)^\top] = B [\text{Some Matrix}] B^\top$$

Crazy simple: $\text{eig}(H) \rightarrow \text{span}(B)$. But this requires gradients of u ...

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-  [Fukumizu 2014]: Gradient Kernel Dimension Reduction (gKDR)

$$\text{replace } \nabla u(X) \text{ with } \nabla f^N(X)$$

See how to optimize the kernel (*i.e.* B) using gradients of $f^N(X)$



`gkdr_demo.ipynb`